

Distinct Values Queries

Time limit: 1.00 s

Memory limit: 512 MB

You are given an array of n integers and q queries of the form: how many distinct values are there in a range $[a, b]$?

Input

The first input line has two integers n and q : the array size and number of queries.

The next line has n integers $x_1, x_2, \dots, x_n$: the array values.

Finally, there are q lines describing the queries. Each line has two integers a and b .

Output

For each query, print the number of distinct values in the range.

Constraints

- $1 \leq n, q \leq 2 \cdot 10^5$
- $1 \leq x_i \leq 10^9$
- $1 \leq a \leq b \leq n$

Example

Input	Output
5 3 3 2 3 1 2 1 3 2 4 1 5	2 3 3